FARUMASI Pharmacist Assistant: A Domain-Specific Question-Answering System for Community Pharmacy in Rwanda

Emmanuel Nsabagasani
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African Leadership University

Project Links

GitHub Repository:
https://github.com/Emmanuel-NS/FARUMASI-AI

Demo Video:
https://drive.google.com/file/d/16IwvICqKY8QzQOUYAPXbfEmvHhQu8fOS/view?usp=sharing

Live System:
https://huggingface.co/spaces/Emmanuel-NS/farumasi-pharmacist-qa

The live system is publicly available on Hugging Face Spaces with ZeroGPU at the time of submission. Assessors should sign in to Hugging Face so free GPU quota is not truncated during a long demonstration session.

1. Abstract

This report presents a domain-specific English question-answering (QA) system for routine medicine and minor-ailment questions in a Rwandan community pharmacy setting. The work is motivated by FARUMASI, a digital pharmacy platform under development by the author, and by the observation that pharmacists spend substantial time answering repeated informational questions while also dispensing and counselling. The system answers only from trusted leaflet text drawn from US openFDA labels for WHO essential medicines [18], [19], MedQuAD [3], MedicationQA [4], and Rwanda FDA patient information leaflets [20]. The implemented pipeline combines a DistilBERT intent classifier [15], a fine-tuned MiniLM dense retriever [6], and a LoRA-adapted Qwen2.5-1.5B-Instruct chat reader [9], [10], with a Flan-T5 LoRA reader [8] retained as a fallback. Against TF-IDF, BM25, zero-shot Transformer, and extractive baselines, LoRA fine-tuning raised gold-context ROUGE-L [16] from 0.327 to 0.643, passage support from 0.62 to 0.96, and abstention on unrelated passages from 25 percent to 73 percent. The final system is deployed as a Gradio web application. The main conclusion is that domain fine-tuning and consumer-style training data improve groundedness and safe refusal more than formal in-domain test scores alone would suggest.

Keywords: domain-specific QA, retrieval-augmented generation, LoRA, biomedical NLP, pharmacy, Rwanda, FARUMASI

2. Introduction

Community pharmacies in Rwanda are often a first access point for medicine advice. Pharmacists are trained to answer patient questions, but they are also responsible for dispensing, stock management, and counselling. In practice, many questions that reach the counter are informational and repetitive: what a medicine is for, how it should be taken, or what side effects to expect. At the same time, patients increasingly look for answers online, where general-purpose chatbots may invent doses, interactions, or contraindications [1], [2].

The motivation for this project comes from FARUMASI, a digital pharmacy marketplace and patient-facing platform that the author has been building for Rwanda. FARUMASI already helps patients find medicines and connect with pharmacies. What it still lacked was a grounded assistant that

can answer routine questions from trusted leaflet text before those questions consume pharmacist time. If such a tool works well, it can reduce the volume of repeated questions directed at busy pharmacies and give patients a safer first stop than informal social media advice.

Language and task selection follow from data availability and user practice. The task is domain-specific open-domain QA over a fixed pharmacy corpus, not machine translation. English was selected because most usable official labels and health-topic resources used for training are written in English [3], [4], [19], while many Rwanda FDA leaflets remain scanned images without a reliable text layer [20]. Rwandan patients nonetheless often use English brand and generic names, so the system expands local lay phrases before retrieval.

Objective. The objective of this project is to design, train, evaluate, and deploy an English-language pharmacy QA assistant that (i) answers routine medicine and minor-ailment questions only from trusted passages, (ii) refuses out-of-scope or unsupported queries, (iii) flags higher-risk situations such as emergencies, and (iv) does so without relying on an external commercial chatbot API.

Scope of contribution. Existing public datasets, pretrained models, and libraries are credited in Sections 3 to 5 and in the reference list. The work implemented by the author includes the pharmacy-specific data preparation pipeline, intent and retrieval fine-tuning, LoRA reader training, query expansion and safety rules, experimental comparison, error analysis, and the Gradio deployment.

3. Related Work

3.1 Biomedical and consumer health QA

Biomedical QA has progressed from classical information retrieval over curated corpora toward neural readers. MedQuAD aggregates large NIH and MedlinePlus-style collections and frames answering partly through question entailment [3]. MedicationQA focuses on real consumer medicine questions answered from trusted resources, which is closer to patient phrasing than clinician notes [4]. Earlier clinical QA often assumed expert users. Consumer health QA must handle lay synonyms, incomplete questions, and higher stakes when the system is wrong [2]. Commercial clinical chatbots and closed APIs are not used here: the assessment requires a trained system rather than a wrapped chatbot, and auditable grounding requires answers that cite passages from a known corpus.

3.2 Retrieval and grounded generation

Dense bi-encoders such as Sentence-BERT and MiniLM map questions and passages into a shared vector space for cosine ranking [5], [6]. Classical BM25 remains strong when rare brand tokens dominate matching [7], which is common for local leaflet names. Retrieval-augmented generation (RAG) conditions a generator on retrieved documents to reduce hallucination [11]. Instruction-tuned generators including Flan-T5 and Qwen2.5 can be adapted with LoRA, which updates low-rank matrices instead of all weights [8], [9], [10]. Prior work shows that fluent generation alone does not guarantee factual grounding [1], [2]. This project connects to that literature by measuring both lexical overlap and a simple support proxy for groundedness, and by training explicit abstention examples.

3.3 Scope control and the gap addressed

Out-of-scope detection for dialogue systems is studied in CLINC150 [13]. Many biomedical QA demos assume in-domain questions and do not combine local Rwandan brand handling, honest refusal for unknown products, and emergency referral to a national number such as SAMU 912. The gap addressed here is therefore practical and technical: a closed-corpus, retrieval-grounded pharmacy assistant adapted to Rwandan community practice and to the workload problem observed through FARUMASI, rather

than another unconstrained chat model.

4. Dataset and Data Preparation

4.1 Sources, size, and characteristics

All training and evaluation text is human-written. No model-generated passages were used as gold answers. The passage corpus contains 8,840 passages. The QA set contains 9,518 question-answer pairs. Pairs are grouped by medicine or condition so the same entity does not appear in more than one split (7,200 train / 1,136 validation / 1,182 test). The mixture is appropriate for the task because it combines trusted regulatory wording, consumer-style questions, out-of-domain negatives, and a smaller set of Rwanda leaflets with local brand names.

Dataset	Role in this project	Size used	Licence or notes
openFDA drug labels [19]	Passages for WHO essential medicines [18]	2,049 passages; 2,523 templated QA pairs	Public domain
MedQuAD [3]	Condition and health-topic QA	5,546 passages; 5,580 QA pairs	CC BY 4.0
MedicationQA [4]	Real consumer medicine questions	655 QA pairs	CC BY 4.0
Rwanda FDA leaflets [20]	Local brands such as Coartem	187 usable leaflets; 590 passages; 760 QA	Public regulatory PDFs
CLINC150 [13]	Out-of-domain intent negatives	23,700 queries (sampled)	CC BY 3.0
ChatDoctor-HealthCare Magic [14]	Consumer questions for intent training only	3,416 questions	Apache 2.0
Challenge set (author)	Informal evaluation questions	56 questions	Author-labelled

Table 1. Datasets used in the project.

4.2 Preprocessing and challenges

- Label sections were cleaned and split into passages of at most 160 words (mean length about 118 words).
- openFDA and Rwanda leaflet questions are templated over real section text. MedicationQA and MedQuAD questions are natural.
- Splits are entity-grouped to avoid leakage across train and test.
- Of 416 cached Rwanda FDA PDFs, 213 scanned images without a text layer were excluded. Two-column layouts sometimes interleaved columns.
- Supplementary corrected labels for metformin, levonorgestrel, and ulipristal replaced wrong combination-product downloads from openFDA.
- Cross-references such as see Warnings and US MedWatch boilerplate were removed before the reader saw the text.

These challenges matter because they affect ecological validity. Templated questions inflate lexical overlap. Clinician-facing FDA wording is harder for patients to read. Missing local brands force honest abstention rather than invention. Those limits motivate the grounded generation design described next.

5. Methodology

The system is a pipeline. A question first passes optional greeting and feedback rules, then a rule-based risk detector, then a learned intent classifier. Lay and brand terms are expanded. A dense retriever ranks passages and applies medicine and section re-ranking. A chat reader generates an answer from on-topic

passages only. Finally, safety localisation replaces US poison numbers with 912 where needed, and the interface adds a disclaimer and source list.

Existing resources versus implemented work. Pretrained checkpoints DistilBERT [15], all-MiniLM-L6-v2 [6], Flan-T5-base [8], and Qwen2.5-1.5B-Instruct [9] are existing models. LoRA [10], Gradio, Hugging Face Transformers, and PEFT are existing libraries. The author implemented data preparation, fine-tuning scripts, query expansion, safety rules, evaluation harnesses, and the deployed application logic.

5.1 Baseline approaches

Intent baseline: TF-IDF features with logistic regression, using word n-grams and word-plus-character n-grams. Retrieval baseline: BM25 [7] and zero-shot MiniLM without domain fine-tuning. Generation baselines: extractive first sentences of the top passage, Flan-T5 zero-shot, and Qwen2.5-1.5B-Instruct zero-shot. These baselines isolate whether domain fine-tuning and LoRA adaptation improve over strong classical and zero-shot options.

5.2 Proposed approach

Intent classifier. DistilBERT with a linear head over three labels (medicine information, condition advice, out of domain). Input length is at most 64 tokens. Training used MedQuAD and CLINC material, then a consumer-augmented set mixing HealthCareMagic questions [14] with heuristic labels.

Retriever. all-MiniLM-L6-v2 maps questions and passages to 384-dimensional normalised embeddings ranked by cosine similarity. Fine-tuning used contrastive learning with in-batch negatives for two epochs, batch size 64, learning rate 2e-5. At inference, the top 50 dense hits are re-ranked with boosts when the medicine name matches (+0.15) and when the section matches the asked aspect (+0.08). A hybrid BM25 and dense fusion through Reciprocal Rank Fusion was also tested.

Answer readers. Flan-T5-base with LoRA (rank 16 on query and value projections) was trained for two epochs at learning rate 3e-4. The deployed reader is Qwen2.5-1.5B-Instruct with LoRA rank 16, alpha 32, and dropout 0.05 on attention and MLP projections. Training used one epoch, batch size 2 with gradient accumulation 8, maximum length 1,536, learning rate 2e-4, and loss on answer tokens only. Example types included ordinary grounded answers, multi-section overviews, and about 8 percent no-answer targets with unrelated passages. Hardware for all reported runs was a free Colab T4 GPU.

Non-learned components. Query expansion, follow-up resolution, unknown-product detection, fuzzy brand suggestion, and risk rules are implemented in application code. They are transparent by design and were refined from live failures that offline benchmarks alone did not expose.

Component	Base model	Adaptation	Key hyperparameters
Intent	DistilBERT-base-uncased [15]	Full fine-tune + consumer data	max length 64; 3 labels
Retriever	all-MiniLM-L6-v2 [6]	Contrastive fine-tune	2 epochs; batch 64; lr 2e-5
Reader (legacy)	Flan-T5-base [8]	LoRA rank 16	2 epochs; lr 3e-4; effective batch 16
Reader (deployed)	Qwen2.5-1.5B-Instruct [9]	LoRA rank 16 [10]	1 epoch; lr 2e-4; batch 2x8; answer-token loss

Table 2. Modelling choices for reported results and the live system.

6. Experiments and Results

Experiments were run on Colab T4 GPUs. Metrics are stored as JSON files in the repository results folder so the tables below can be verified against committed artefacts.

6.1 Intent classification

The main research question was whether DistilBERT and consumer-style data outperform TF-IDF on informal questions. Models trained only on formal MedQuAD and CLINC text scored above 99 percent on the held-out formal test split but collapsed on the informal challenge set. That pattern indicates style overfitting rather than true topic understanding, consistent with out-of-scope evaluation concerns raised around CLINC150-style settings [13].

Model	Test acc.	MedicationQA acc.	Challenge acc.	Challenge macro-F1
TF-IDF + LR (word)	0.993	0.528	0.393	0.372
TF-IDF + LR (word + char)	0.995	0.694	0.464	0.420
TF-IDF + LR + patient questions	0.979	0.556	0.768	0.764
DistilBERT (original)	0.998	0.917	0.554	0.486
DistilBERT + patient questions (deployed)	0.994	0.889	0.804	0.782

Table 3. Intent results. Formal-test accuracy is misleading. Challenge-set accuracy is the more realistic criterion, and consumer augmentation is the decisive factor.

6.2 Retrieval

Fine-tuning MiniLM more than doubled BM25 Recall at 1. Adding Rwanda leaflet questions helped only about one point on Rwanda Recall at 1, which is within noise for n equals 180. The model mainly learns a question-to-section mapping that transfers across sources. BM25 remains competitive on rare brand tokens, as expected from classical IR [7].

Model	R@1	R@5	MRR@10	Topic@1	Rwanda R@1
BM25	0.322	0.706	0.478	0.852	0.444
MiniLM zero-shot	0.436	0.804	0.589	0.882	0.400
MiniLM fine-tuned (no Rwanda)	0.694	0.954	0.809	0.977	0.667
MiniLM fine-tuned (all, deployed)	0.695	0.958	0.810	0.977	0.678
Hybrid RRF	0.551	0.867	0.684	0.951	0.689

Table 4. Retrieval on 1,182 test questions and 8,840 passages.

6.3 Answer generation

Gold context isolates the reader. Retrieved context tests end-to-end behaviour with top-3 BM25 passages so Flan and Qwen tables remain comparable. Qwen with LoRA roughly doubles zero-shot ROUGE-L and sharply improves support and abstention. That result supports RAG plus parameter-efficient adaptation [10], [11] over prompting a base model alone.

Reader	Context	ROUGE-L	BERTScore	Support	Abstain unrelated
Extractive*	gold	0.931*	0.985*	-	no
Flan-T5 zero-shot	gold	0.191	0.761	-	no
Flan-T5 + LoRA	gold	0.608	0.900	-	no
Qwen zero-shot	gold	0.327	0.829	0.624	25%

Reader	Context	ROUGE-L	BERTScore	Support	Abstain unrelated
Qwen + LoRA (deployed)	gold	0.643	0.911	0.964	73%
Extractive	retrieved	0.395	0.833	-	no
Flan-T5 + LoRA	retrieved	0.414	0.841	-	no
Qwen + LoRA	retrieved	0.432	0.847	0.930	73%

Table 5. Generation results. The starred extractive gold score is an upper bound because references are the first sentences of the gold passage. Support is the share of answer content words found in the given passages. BERTScore follows Zhang et al. [17].

On the labelled challenge risk set, the rule-based detector reached accuracy 1.0 across routine, prescription, special-group, and emergency classes. This score is optimistic because the same set helped refine the rules. It is reported for transparency, not as independent clinical validation.

7. Error Analysis and Discussion

7.1 Metrics

Accuracy and macro-F1 are used for intent because classes are imbalanced. Recall at k and MRR at 10 are standard IR metrics for retrieval [7]. Topic at 1 checks whether the right medicine is found even if the wrong section is ranked first. ROUGE-L [16] and BERTScore [17] measure answer overlap and semantic similarity. Support and abstention capture groundedness and safe refusal. None of these metrics equal clinical correctness.

7.2 Successful examples

- What are the side effects of metformin? The system lists effects from the single-ingredient label rather than an earlier wrong combination product.
- after sex pills and a follow-up about 120 hours map local phrasing to levonorgestrel and then ulipristal.
- quarteme is treated as a near miss for Coartem, with a malaria overview for artemether and lumefantrine.
- Exchange-rate questions are refused as out of scope.

7.3 Failure modes and interpretation

- Style overfitting in intent: without consumer questions, DistilBERT was near perfect on formal text but only about 55 percent accurate on the challenge set.
- Aspect errors in retrieval: BM25 Topic at 1 is high (0.85) while Recall at 1 is low (0.32), so the drug is found but the wrong section is ranked first.
- Wrong openFDA products: richest-label downloads sometimes selected combination products until supplementary labels were added.
- Unknown local brands: Kopavitamin initially triggered generic vitamin text until a name checker forced an honest not-in-sources reply.
- Opener mismatch: the reader sometimes announced the wrong section for follow-ups; a post-hoc opener correction aligns the heading with the question.
- Loose abstention: on deliberately unrelated passages, Qwen LoRA still answers about 27 percent of the time, so a retrieval similarity threshold remains a second guard.

Discussion. Three findings align with prior work. First, high in-domain accuracy can hide poor generalisation to consumer style [13]. Second, sparse and dense retrievers trade off rare-token matching against semantic matching [5], [7]. Third, zero-shot instruction models are fluent but under-grounded, while LoRA on grounded targets restores fidelity [1], [10], [11]. Higher abstention matters for

FARUMASI: refusing an unknown brand is safer than inventing ingredients and more useful to a busy pharmacist.

8. Deployment

The final model is deployed as a Gradio 5 web application on Hugging Face Spaces with ZeroGPU. At startup, DistilBERT, MiniLM embeddings, and the Qwen base model with its merged LoRA adapter load on CPU. A GPU wrapper moves compute to the GPU only for each answer, typically for about five to ten seconds. Users type a question in the chat interface and receive an answer, an Understood as line for follow-up interpretation, model component tags, source passages with similarity scores, and a disclaimer.

Live System: https://huggingface.co/spaces/Emmanuel-NS/farumasi-pharmacist-qa

Demo Video: https://drive.google.com/file/d/16IwvICqKY8QzQOUYAPXbfEmvHhQu8fOS/view?usp=sharing

The recorded demonstration shows the complete user workflow, successful and unsuccessful cases, and how the web application connects to the trained models. From a FARUMASI perspective, deployment matters because the assistant is intended to sit in front of real users and reduce repeated informational load on pharmacies, not to replace professional care.

9. Limitations and Future Work

9.1 Limitations

- English only. Kinyarwanda and French remain out of scope for this version.
- Templated openFDA and Rwanda questions, and clinician-facing label wording, limit ecological validity.
- ROUGE and BERTScore are not clinical correctness. No clinician adjudication was performed.
- Challenge and safety sets are small. Safety accuracy is optimistic.
- Local brands absent from leaflets cannot be described beyond honest abstention.
- ZeroGPU free quota can interrupt anonymous sessions.
- The assistant is not medical advice. Emergencies require SAMU 912 or a pharmacist.

9.2 Future work

With more time, data, and compute, the main improvements would be multilingual support with OCR for scanned Rwanda leaflets, clinician evaluation of answer safety, larger informal challenge sets, tighter integration into FARUMASI patient flows, and stronger abstention calibration so loosely related passages do not produce unsupported answers.

AI assistance note. AI coding tools helped scaffold code and draft some challenge items. All training, evaluation, and live behaviour were run and verified by the author, who takes full responsibility for the submission. Repository commits are authored as Emmanuel-NS.

10. Conclusion

This project delivered a complete domain-specific pharmacy QA system grounded in the practical needs of FARUMASI and of busy community pharmacies in Rwanda. Using trusted public datasets and leaflets [3], [4], [19], [20], the author fine-tuned DistilBERT for intent control, MiniLM for retrieval, and a LoRA-adapted Qwen2.5 reader for grounded answers, then deployed the system on the web. The experiments show that domain fine-tuning and consumer-style data matter more than raw formal-test

scores, that LoRA roughly doubles zero-shot Qwen answer quality while improving groundedness and abstention, and that live testing surfaces failure modes that offline benchmarks miss.

What was learned is that a useful pharmacy assistant is not only a fluent generator. It is a system that retrieves the right leaflet text, stays faithful to that text, refuses when sources are missing, and leaves clinical responsibility with pharmacists and emergency services. That is the design direction needed if FARUMASI is to reduce repeated questions at the counter without creating new safety risks.

11. References

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